Topic 4: Exchange and Transport

This topic considers the requirements for transport mechanisms in cells and mass flow systems in organisms. The roles of the components of the mammalian circulatory system and the vascular system in plants are studied. Practical skills are developed through the investigation of factors that affect membrane permeability and water potential of plant tissues.

Opportunities for developing mathematical skills within this topic include: recognising and making use of appropriate units in calculations; recognising and using expressions in decimal and standard form; using ratios, fractions and percentages; finding arithmetic means; solving algebraic equations; translating information between graphical, numerical and algebraic forms; plotting variables from experimental data; understanding that $y = mx + c$ represents a linear relationship; determining the intercept of a graph; calculating rate of change from a graph showing a linear relationship; calculating the circumferences, surface areas and volumes of regular shapes. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:	
4.1	**Surface area to volume ratio**
i	Understand how surface area to volume ratio affects transport of molecules in living organisms.
ii	Understand why organisms need a mass transport system and specialised gas exchange surfaces as they increase in size.